

Frozen-Expert Mixture of Experts

for Code-Mixed Language Modeling in Indian Social Media

Training only ~200K parameters to match 85M-parameter fine-tuning

CS F425 · Deep Learning · BITS Pilani

Team

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Problem & Motivation

Hindi-English code-mixing is everywhere - NLP models struggle

💬 *"Yaar yeh movie was actually amazing, bilkul waste of time nahi tha."*

⚠️ Data Scarcity

- Few large annotated code-mixed corpora
- Most are noisy Twitter scrapes
- Inconsistent romanisation

⚠️ Spelling Variation

- Same Hindi word, many spellings
 - "nahi" / "nahin" / "nhi"
- Vocabulary-based models break

⚠️ Dual-Language Demands

- Model must understand Hindi morphology
- AND English semantics simultaneously
- Per-token language identity required

Why not just fine-tune a multilingual model?

- Updates ~85M parameters - expensive GPU memory & time
- Risks catastrophic forgetting of pre-trained Hindi/English knowledge
- Monolithic: forces one set of weights to handle both languages simultaneously

Our Approach - FE-MoE

Frozen-Expert Mixture of Experts · only the router is trained

Keep TWO complementary frozen experts. Train ONLY a lightweight router (~200K params) that learns per-token how much to trust each expert.

❄ HingBERT (FROZEN)

Hindi-English Specialist
52.9M tweets pre-training
WordPiece tokeniser



❄ RoBERTa-base (FROZEN)

English Generalist
RoBERTa pre-training
BPE tokeniser



Learned Router → $\alpha \in (0,1)$ per token

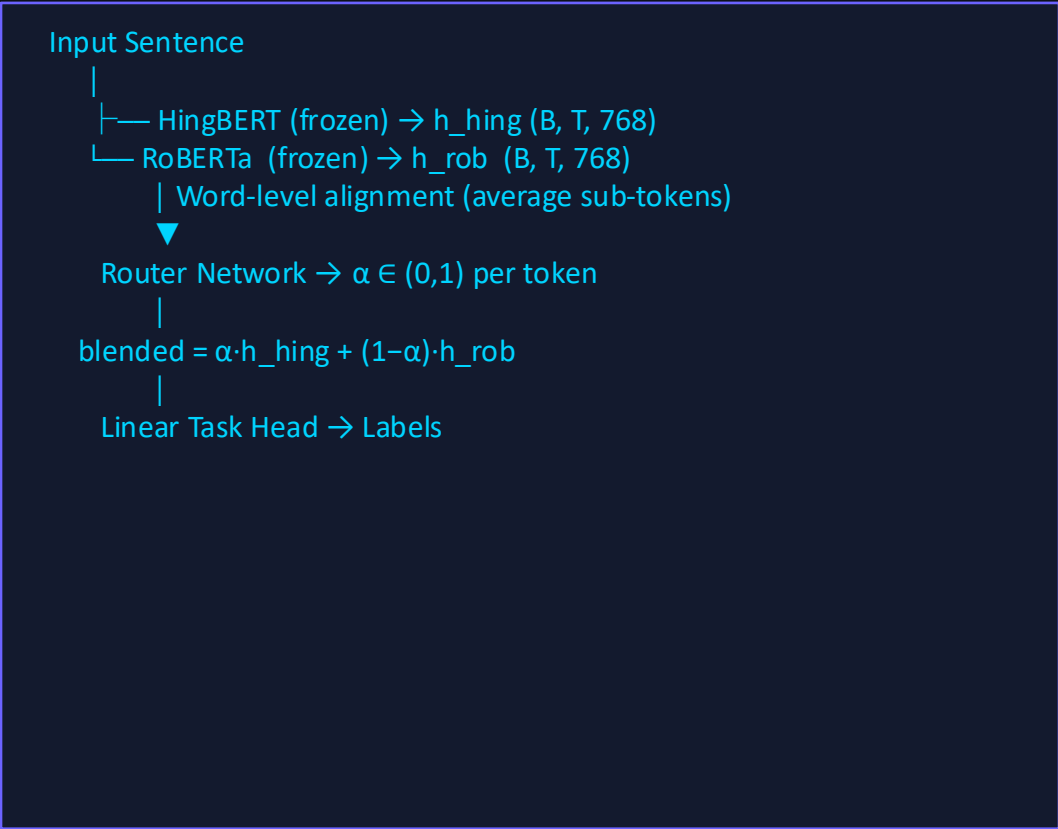
$$blended = \alpha \cdot h_{HingBERT} + (1-\alpha) \cdot h_{RoBERTa}$$

Key Advantages

- ✓ 425× fewer trainable params than B1
- ✓ No catastrophic forgetting
- ✓ Language-specific specialization
- ✓ Implicit routing - no LID labels needed
- ✓ Three router architectures explored

Architecture Deep Dive

Sub-token alignment + three learned router architectures



RouterMLP

R1-R7

- Linear(1536→256) → GELU
- Linear(256→1) → $\sigma(\cdot/\tau)$
- ~200K params
- Token-level routing

RouterGRU

R8

- BiGRU(hidden=128)
- Linear(256→1) → $\sigma(\cdot/\tau)$
- Sequential context
- Past + future tokens

RouterCNN

R9

- Conv1d(k=5, 128 ch)
- Linear(128→1) → $\sigma(\cdot/\tau)$
- Local 5-token context
- Cheaper than GRU

Sub-Token Alignment

WordPiece (HingBERT) & BPE (RoBERTa) split words differently.
Average all sub-tokens per word → consistent (B, T, 768) tensors.

Routing Modes Explored

Soft token-level (default) | Hard routing / STE (R7) | Sentence-level CLS (R6)

Datasets & Tasks

COMI-LINGUA - expert-annotated Hindi-English code-mixed corpus

COMI-LINGUA Dataset

- Expert-annotated social media text (romanised)
- High inter-annotator agreement ($\kappa \geq 0.81$)
- Native train/val/test splits (10% train $\rightarrow$ validation)
- Devanagari sentences filtered - purely romanised/English code-mixed

Train / Val / Test - LID & POS: native dataset split, 10% train $\rightarrow$ val (seed-controlled)
All datasets downloaded automatically at runtime via HuggingFace Hub

Training Configuration

- AdamW lr=2e-4 | Batch size 32 | Max seq len 128
- Linear warmup 10% | Gradient clip 1.0 | Max 20 epochs
- Early stopping patience 3 | Best-weight restore
- 2 seeds {42, 123} $\rightarrow$ 26 trained models across 13 configs

Language ID (LID)

hi / en / ot labels per token
Most fundamental code-mixed task

Metric: Weighted F1

POS Tagging

14 universal POS labels
Reveals syntactic structure of mixed sentences

Metric: Token Accuracy

NER (explored)

BIO: PERSON, ORG, LOC
Near-zero F1 with frozen experts
- excluded from main sweep

Metric: seqeval entity F1

Experiments

4 baselines · 9 MoE router configs · 2 ablations = 15 configurations × 2 seeds

Baselines

ID	Model	Frozen?	Trainable Params
B1	HingBERT fine-tuned	❌ No	~85M
B2	HingBERT frozen + head	✅ Yes	~768×N
B3	RoBERTa frozen + head	✅ Yes	~768×N
B4	Fixed 50/50 avg + head	✅ Yes	~768×N

Router Input Ablations

ID	Router Input	Notes
R1 (A1)	Both (concat 1536-d)	Baseline ablation
A2	HingBERT only (768-d)	Router sees only h_hing
A3	RoBERTa only (768-d)	Router sees only h_rob

MoE Router Variants

ID	Router	τ	Mode
R1	MLP	1.0	Soft token
R2	MLP	0.1	Soft token ★
R3	MLP	0.3	Soft token
R4	MLP	0.5	Soft token
R5	MLP	2.0	Soft token
R6	MLP	1.0	Sentence-level
R7	MLP	1.0	Hard (STE)
R8	BiGRU	1.0	Soft token
R9	CNN	1.0	Soft token

★ R2 (τ =0.1) = BEST | Low temperature → sharp near-binary routing → stable training

Main Results

Mean $\pm$ std over two seeds (42, 123) | 🌟 = best · ★ = best MoE

88.98%

Best MoE LID

R2 · $\tau=0.1$

89.65%

Fine-tuned (B1)

Full 85M-param upper bound

0.67 pts

Gap to B1

LID F1 difference

425×

Param Savings

200K vs 85M

ID	Description	LID F1 (%)	POS Acc (%)
B1 🌟	HingBERT fine-tuned	89.65 $\pm$ 0.23	91.09 $\pm$ 0.11
B2	HingBERT frozen	80.55 $\pm$ 0.06	65.23 $\pm$ 0.25
B3	RoBERTa frozen	81.48 $\pm$ 0.01	63.18 $\pm$ 0.31
B4	Fixed 50/50 avg	80.41 $\pm$ 0.31	61.10 $\pm$ 0.12
R2 ★	MoE MLP $\tau=0.1$	88.98 $\pm$ 0.65	66.99 $\pm$ 1.54
R3	MoE MLP $\tau=0.3$	88.86 $\pm$ 0.29	66.99 $\pm$ 1.25
R9	MoE CNN k=5	88.29 $\pm$ 0.12	66.78 $\pm$ 0.01
R7	MoE MLP hard routing	87.58 $\pm$ 0.10	65.20 $\pm$ 0.22
R8	MoE BiGRU router	86.61 $\pm$ 0.63	64.22 $\pm$ 2.17
R6	MoE sentence-level	81.20 $\pm$ 0.04	66.44 $\pm$ 0.18

Key Findings

What the numbers tell us about frozen-expert routing

200K params $\approx$ 85M params on LID

Best MoE (R2) reaches 88.98% F1 - just 0.67 pts below fully fine-tuned B1 (89.65%).

Training cost: 0.24% as many parameters.

Temperature matters - lower is better

$\tau=0.1$ (near-hard routing) gives best LID and lowest variance ($\pm 0.65\%$).

$\tau=1.0$ and $\tau=2.0$ show high variance ($\pm 3.2\%$) - unstable non-convex landscape.

CNN $\approx$ MLP on LID | BiGRU underperforms

CNN (R9: 88.29%) nearly matches MLP (R2: 88.98%) at lower sequential cost.
BiGRU (R8: 86.61%) - error propagation across language-switch boundaries hurts.

Per-token routing is essential

Sentence-level routing (R6: 81.2%) degrades to near frozen-baseline.
Code-mixing alternates at word level - one α per sentence cannot cope.

POS gap reveals frozen-expert limits

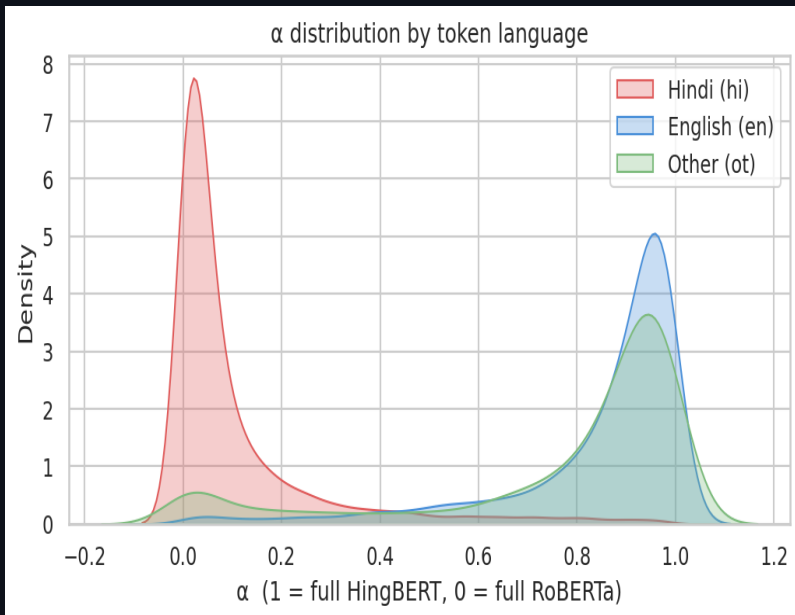
MoE plateaus at $\sim 67\%$ POS accuracy vs B1's 91.09%.

Frozen MLM pre-training lacks POS-specific geometric structure.

Interpretability: What the Router Learned

Language-aware routing emerges without any explicit LID supervision

The router discovers language identity from frozen hidden-state geometry alone - no language labels are ever provided during training.



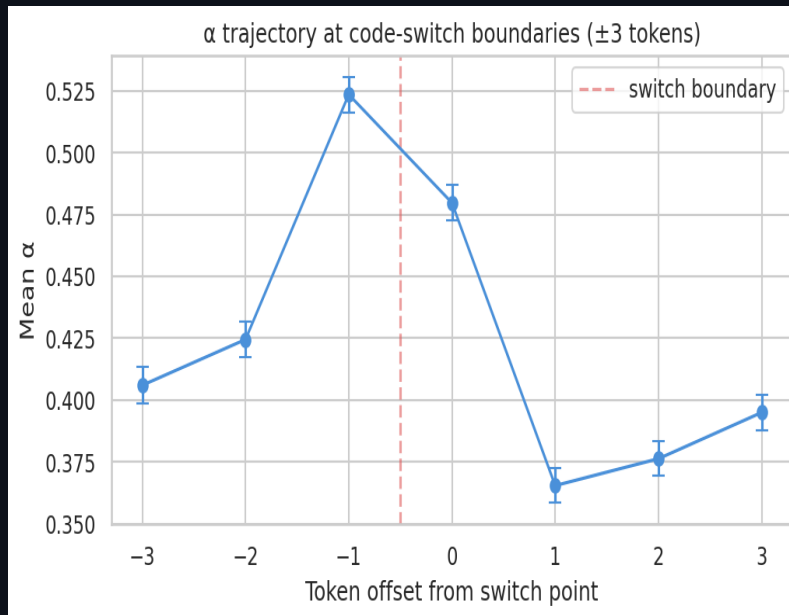
α by Token Language

Hindi tokens → higher α (more HingBERT)

English tokens → lower α (more RoBERTa)

Other tokens: in between

Emerges WITHOUT any α supervision

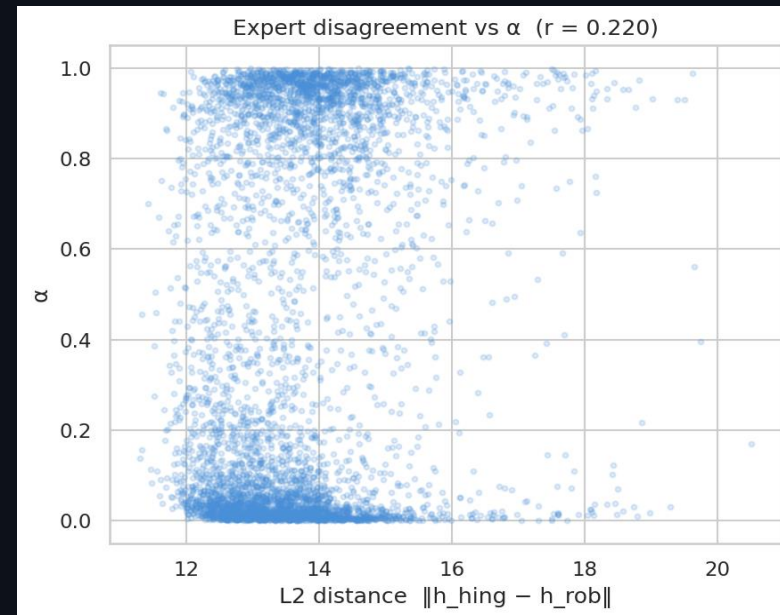


α at Code-Switch Boundaries

Smooth α transition at language switch

Inflection at offset 0 (switch point)

Router tracks language context
across multiple tokens



Expert Disagreement vs α

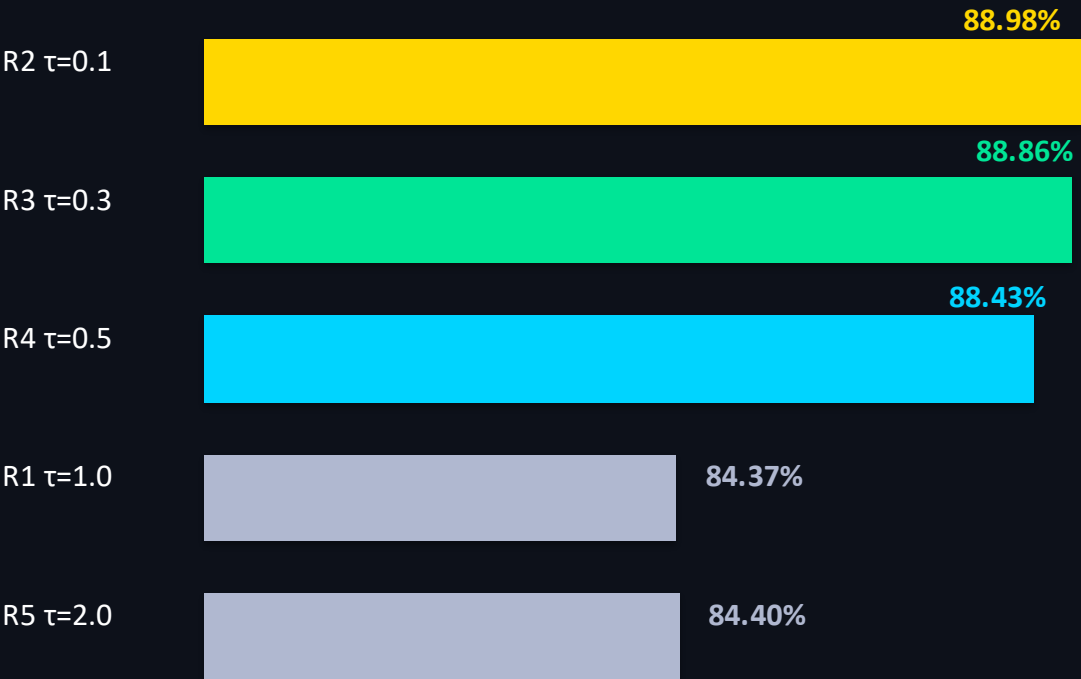
Higher $\|h_{\text{hing}} - h_{\text{rob}}\|_2 \rightarrow$ more
decisive routing ($\alpha \rightarrow 0$ or 1)

Router uses expert disagreement
as routing signal

Ablation Study

Temperature · routing granularity · router input - what matters?

Temperature (τ) Ablation - LID



Routing Granularity Impact

Hard token (R7): 87.58% (+3.21 vs R1)

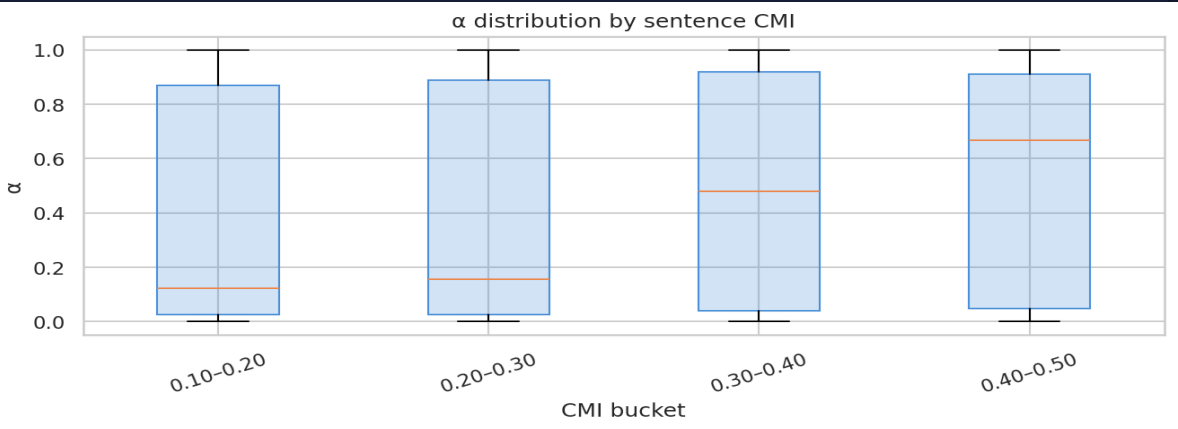
Soft token (R1): 84.37% (Baseline)

Sentence-level (R6): 81.20% (-3.17 vs R1)

Router Input Ablation - LID F1

A3 - RoBERTa only	+3.05
87.42 ± 0.08	
A2 - HingBERT only	+1.69
86.06 ± 0.24	
R1 - Both (concat)	baseline
84.37 ± 3.21	

Surprising finding:
RoBERTa-only routing (A3) beats both HingBERT-only (A2) and full concat (R1). RoBERTa's compositional English representations encode more discriminative routing structure - even for Hindi tokens.



Limitations & What Didn't Work

Honest analysis of failure modes - key for scientific rigour

✗ MoE on POS Tagging

- MoE plateaus at ~67% vs B1's 91.09% → -24 points gap
- Frozen MLM pre-training lacks POS-specific geometric structure
- Router cannot compensate without reshaping expert representations
- Fix: partially unfreeze top 2 transformer layers

✗ NER with Frozen Experts

- All frozen models achieve near-zero entity F1
- Entity boundary detection requires task-specific fine-tuning
- NER was excluded from the main experiment sweep
- Fix: fine-tune expert token representations for NER

✗ Hard Routing on POS (R7)

- Hard binary gating hurts POS accuracy (65.2%)
- Discrete routing loses gradient signal for task head
- Soft blending is valuable for ambiguous / mixed tokens
- Fix: keep soft routing for POS; use hard only for LID

✗ Sentence-Level Routing (R6)

- One α per sentence → LID degrades to 81.2% (near frozen baselines)
- Code-mixing alternates at word level - single α is insufficient
- Per-token granularity is non-negotiable for code-mixed text
- Fix: always use token-level routing

Conclusion

FE-MoE: Competitive code-mixed NLP at a fraction of the cost

R2 achieves 88.98% LID F1 - within 0.67 points of full 85M-parameter fine-tuning - while training ONLY ~200K parameters (0.24% of B1). A 425× reduction in trainable parameters.

① Sharp routing wins

Low temperature ($\tau=0.1$) produces stable, near-binary routing → best performance and lowest variance across seeds.

② CNN $\approx$ MLP for code-mixing

Local 5-token context (CNN R9) matches MLP performance on LID at lower sequential compute cost.

③ Per-token granularity is non-negotiable

Sentence-level routing (R6) degrades to frozen-baseline performance. Code-mixing alternates at word level - one α per sentence fails.

④ Router learns linguistics implicitly

α separation by language label, smooth switch-point transitions, and disagreement-driven routing all emerge from task loss alone.

Future Work: Partially unfreeze top 2 expert layers to close the POS gap | Extend to NER | Multilingual expert triplets | Sparse top-k routing